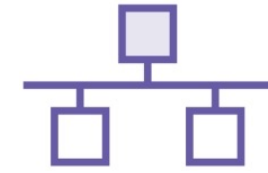
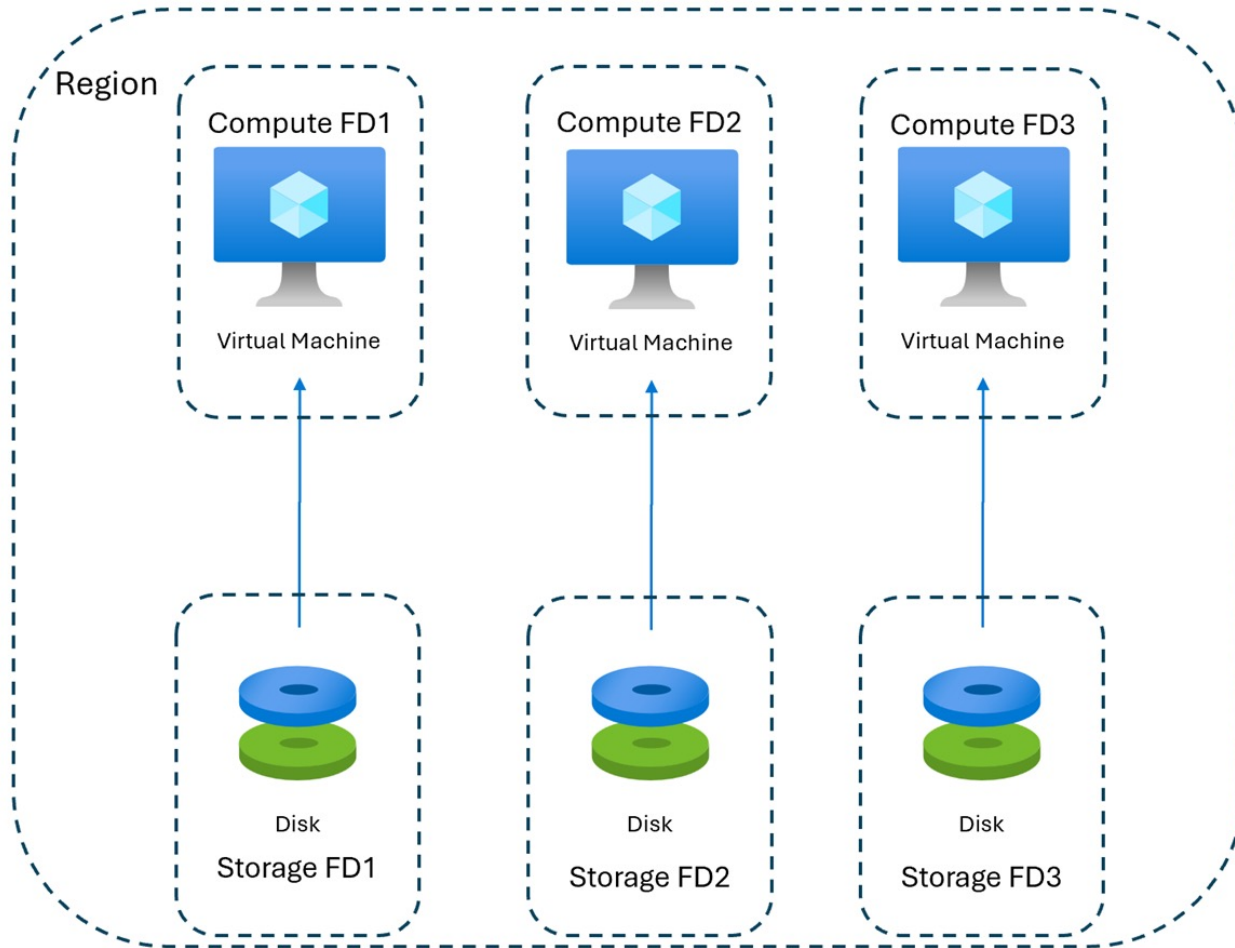


Availability Sets



VMs are placed on nodes in a rack



Failures can occur at node and rack levels

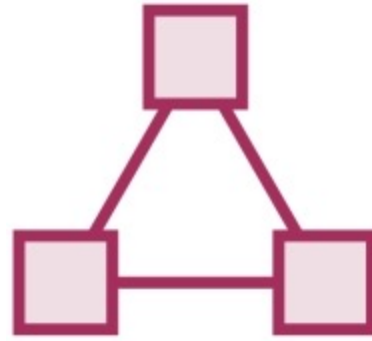


Maintenance is also required on hosts and VMs are **not** live migrated

Availability Sets



To ensure availability of services, always deploy minimum 2 instances of any service and place in a unique availability set

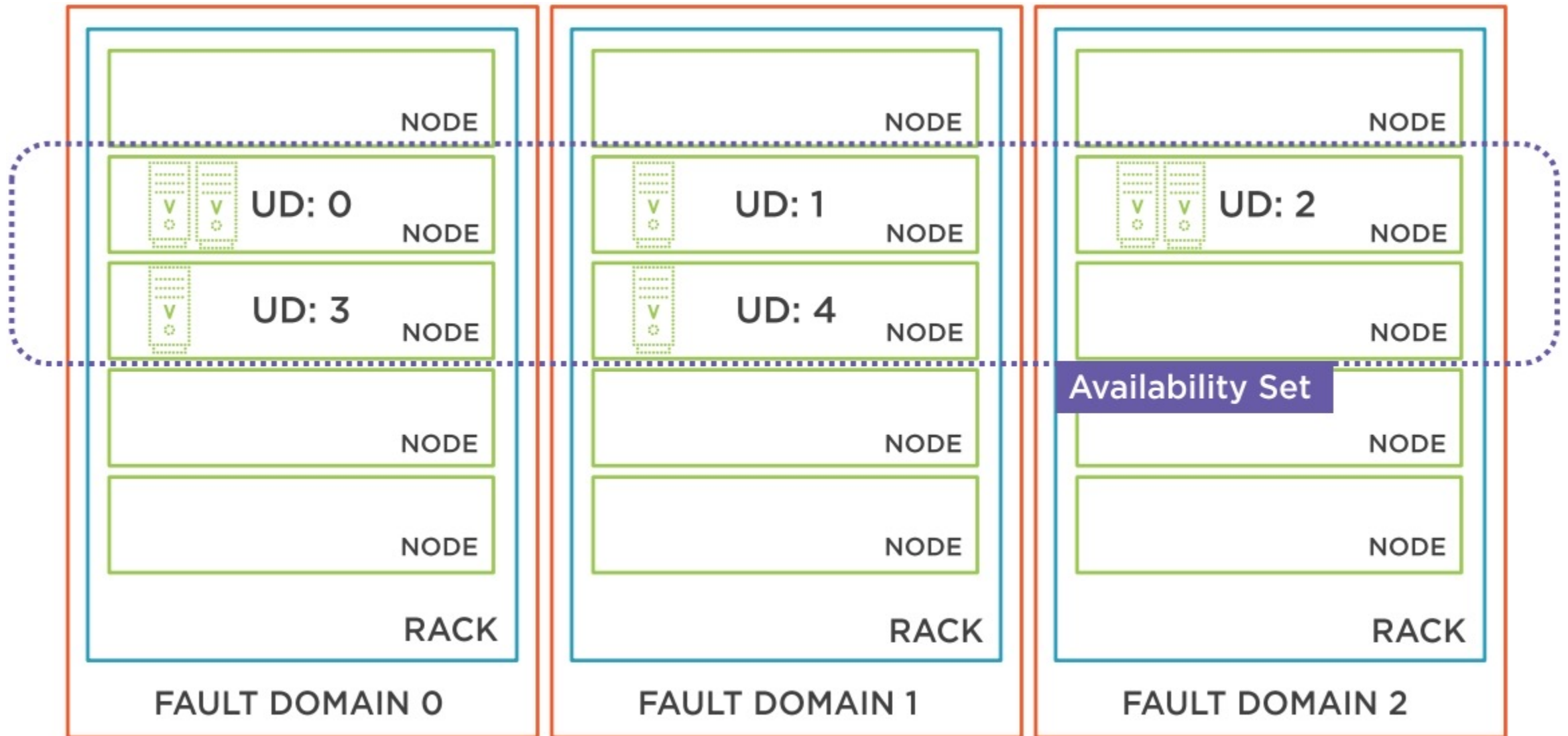


This ensures VMs are spread over three fault domains (racks) and five (by default) update domains



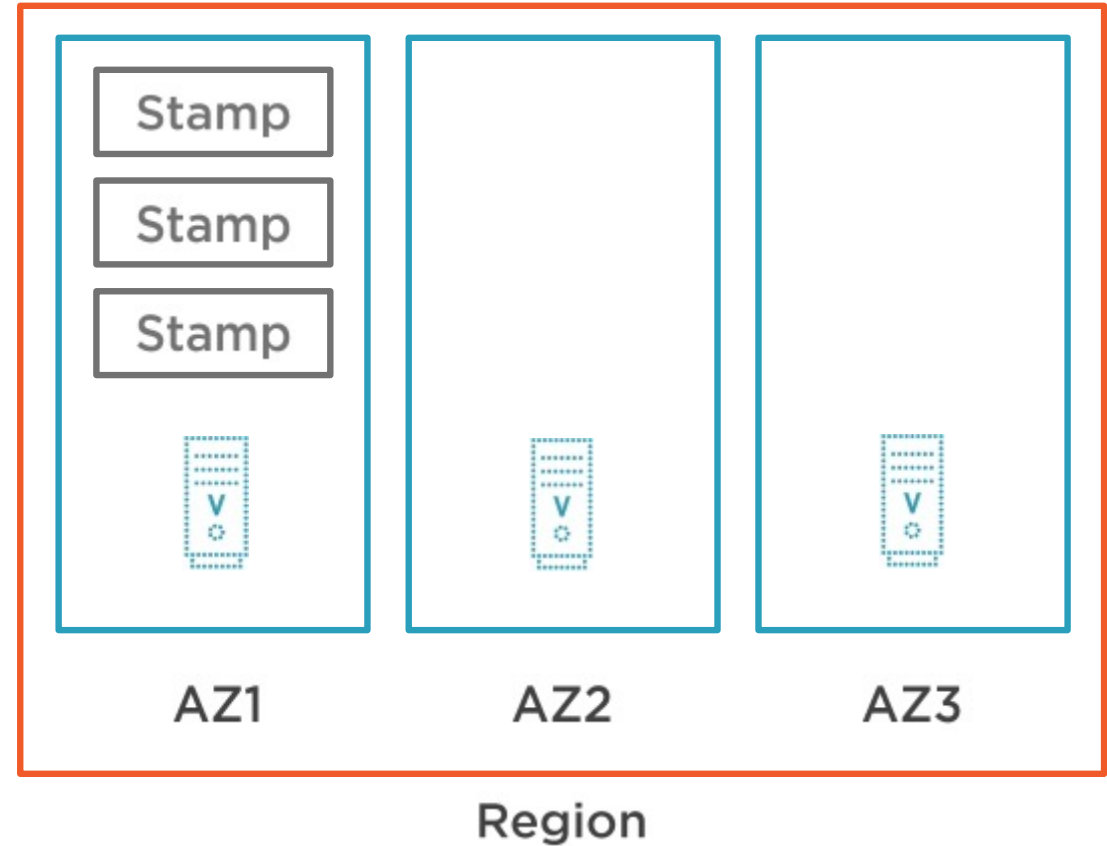
Must have minimum 2 VMs to receive SLA of 99.95%

Availability Sets



Availability zones

- Regions are broken up into physically separate AZs
- AZs have independent power, cooling and networking
- 3 AZs are exposed per subscription
- VMs spread over AZs receive 99.99% SLA
- Each AZ can be thought of as separate fault domain and update domain
- Virtual networks span AZs



Regions

A world map with a dark blue background, showing various regions marked with blue dots and labels. The map includes labels for North America (West US 2, West Central US, West US, US Gov Arizona, South Central US, US Gov Texas, Central US, US Gov Iowa, North Central US, Canada East, Canada Central, US DoD East, East US, East US 2, US Gov Virginia, US DoD Central, Brazil South), Europe (Norway West, Norway East, West Europe, UK South, North Europe, UK West, France Central, France South, Germany West Central, Germany North, Germany Northeast, Germany Central, Switzerland North, Switzerland West), Africa (South Africa North, South Africa West), Asia (UAE North, UAE Central, West India, Central India, South India, China North, China North 2, Korea Central, Korea South, Japan East, Japan West, China East 2, China East, East Asia, Southeast Asia), and Australia (Australia East, Australia Central, Australia Central 2, Australia Southeast). A legend in the bottom left corner indicates: a blue dot for 'Available region', a dashed circle for 'Announced region', and a blue diamond for 'Availability Zones'. An orange banner at the bottom right contains the text '60+ regions 140 countries'.

Available region

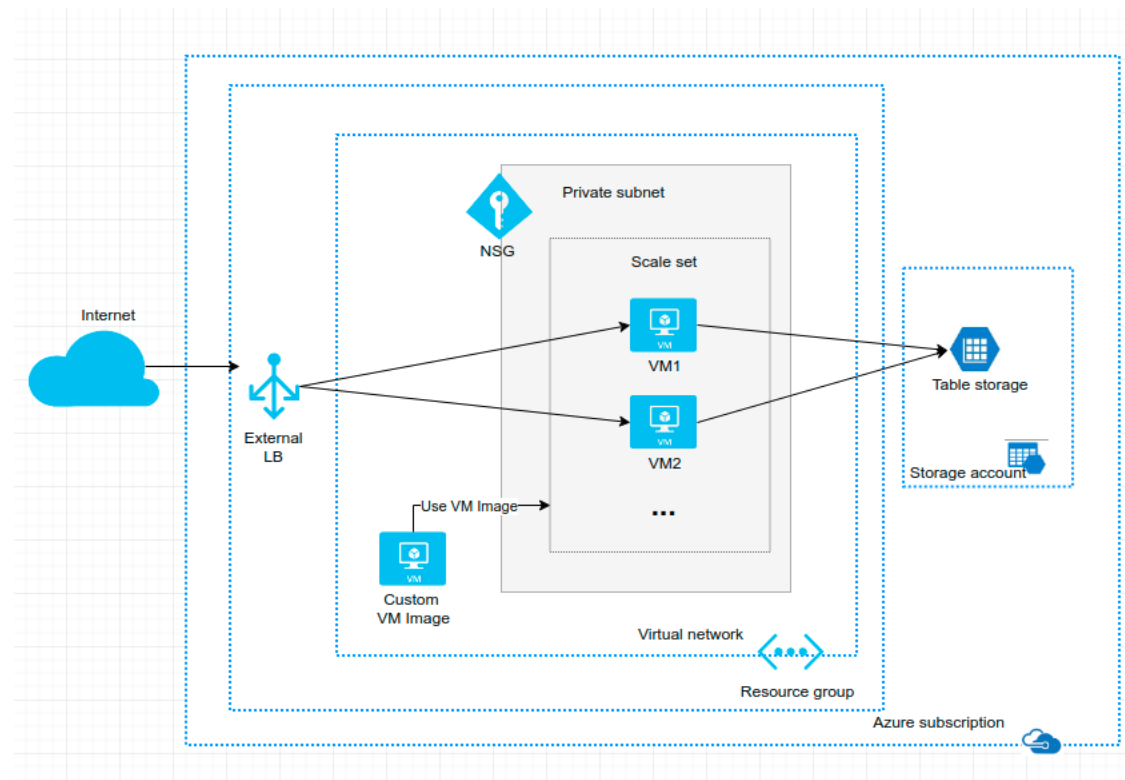
Announced region

Availability Zones

60+ regions 140 countries

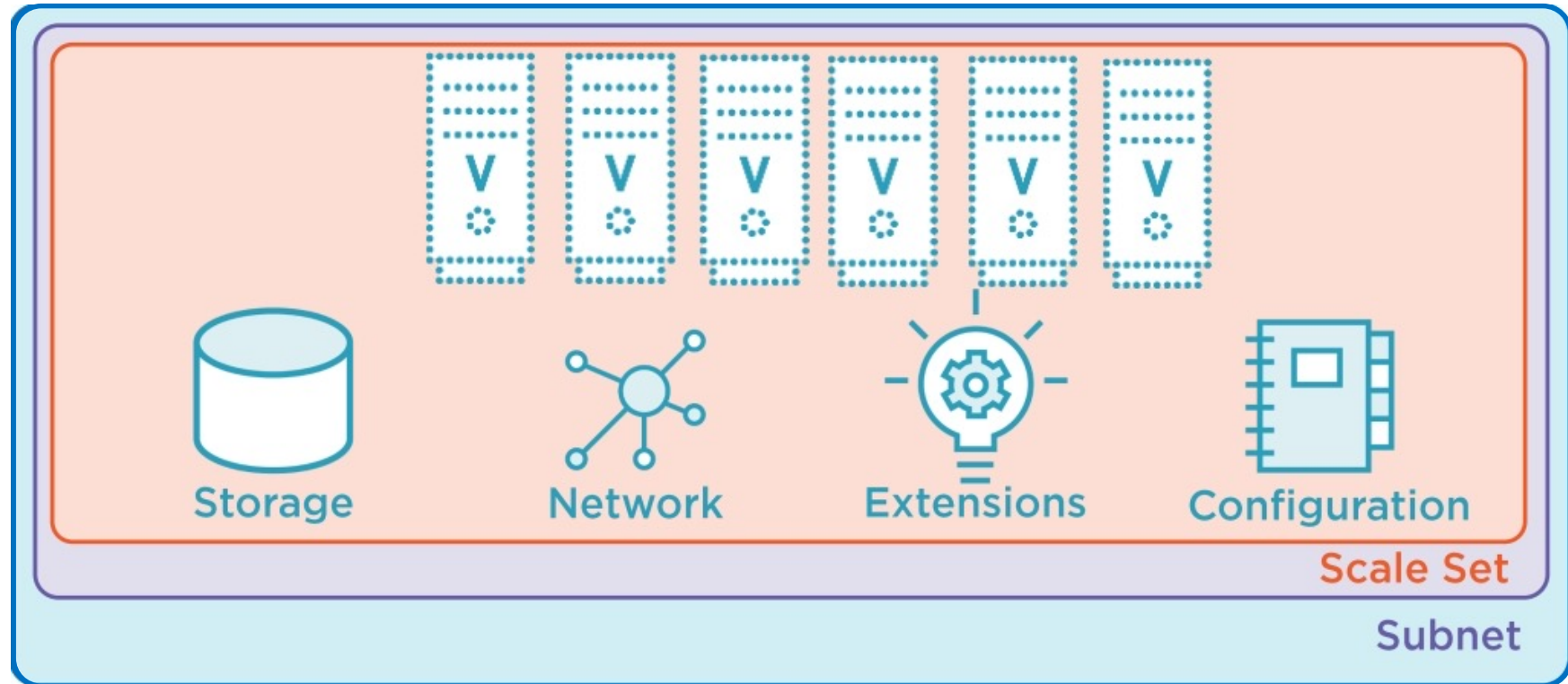
Scale sets

- Enables large scale deployment of VMs from single gold image with automatic configuration
- Supports auto-scale based on metrics and schedules
- Scale sets can be updated without taking down the entire set



Scale sets

- Azure VM Scale Sets enable the entire deployment to be managed in a simple fashion
- Enable rollout of updates without taking down the entire service.



Scale sets Limitations



- 1000 VMs per Scale Set maximum and use with Azure Standard Load Balancer for matching scale



- 2000 Scale Sets per region per subscription

Auto-scaling Scale Sets

Scheduled

Metric-Based

Scale Sets support many types of scale

- On a schedule both specific day and recurrence (enables scaling ahead of the load increase)
- Based on resource metrics
- Can combine (schedule rules take precedence over metric rules)

Pop quiz:

You need high availability in a single region and want to support 200 instances in a scale set. Which option should you choose?

- A. Regional scale set (non-zonal) without placement groups
- B. Zonal scale set with true Availability Zones
- C. Regional scale set with placement groups
- D. Single VM in an Availability Set

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